

Sajeevan Veeriah

Mechatronics Engineer

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Graduate Member, Engineers Australia

PROFESSIONAL SUMMARY

Mechatronics engineer with hands-on delivery across industrial automation, SCADA and HMI systems, control systems, embedded systems, electronics, mechanical design and applied AI/ML. Recent experience covers smart factory automation for regulated pharmaceutical, biotech and food manufacturing, SCADA platform migration, CAN-based vehicle software and ADAS validation, and end-to-end IoT deployments. Strong at integrating mechanical systems, electrical hardware, sensors, drives, control logic, HMIs and data pipelines into reliable, traceable and testable systems, backed by FAT, SAT, commissioning and qualification discipline.

TECHNICAL SKILLS

Controls and Automation: PLC programming (IEC 61131-3), Siemens TIA Portal, WinCC, PCS 7, iFIX, PVI+, HMI, SCADA, MES, batch execution, Modbus TCP/RTU, Profinet, industrial Ethernet, FAT, SAT, commissioning and troubleshooting.

Robotics and Autonomy: ROS 2 (Humble), Nav2, MoveIt 2, Gazebo, RViz, SLAM, A* path planning, Kalman/EKF state estimation, sensor fusion, LiDAR, IMU, depth cameras, OpenCV, motion control and PID tuning.

Embedded Systems and Electronics: ESP32 (incl. ESP32-S3), STM32, embedded C/C++, FreeRTOS, UART, I2C, SPI, CAN, ADC, PWM, PCB design (Altium, KiCad), schematic capture, component selection, signal conditioning, sensor and actuator integration, and board-level debugging and bring-up.

Mechanical Design and CAD: SolidWorks, Fusion 360, part and assembly modelling, engineering drawings, drawing interpretation, GD&T basics, mechanical integration, tolerance considerations, FEA exposure, 3D printing (FDM), mechanism design, and pneumatic and hydraulic systems.

Electrical and Electromechanical: Control schematics and wiring diagrams, VFDs and motor drives, motor control fundamentals, contactors, relays and switchgear, power distribution, sensors and transducers, actuators, instrumentation wiring, cable routing and AS/NZS 3000 awareness.

Software, Data and AI/ML: Python, C, C++, MATLAB, Simulink, JavaScript, TypeScript, Node.js, REST APIs, MQTT, Git, Linux, JIRA, test automation, InfluxDB, Grafana, data analysis, ML fundamentals and anomaly detection.

Quality and Standards: GAMP 5, FDA 21 CFR Part 11, GMP, ISO 13849, IEC 62443, ADR and EURO standards, ITPs, MDRs, FMEA, QA records, traceability, validation documentation and commissioning and qualification evidence.

PROFESSIONAL EXPERIENCE

Automation Engineer | JAG Process Solutions Pty Ltd

Jan 2026 - Jun 2026

Brunswick, VIC

- Delivered automation and smart factory engineering for pharmaceutical, biotech, food and process clients operating under GMP, validation and traceability requirements.
- Executed a SCADA platform migration from iFIX to PVI+ as a priority delivery item, converting application content and verifying functionality against the existing validated system.
- Developed and maintained control logic, HMIs, process visualisation, MES and batch execution functions across automated plants, skids and package units.
- Produced requirements, functional design, FAT, SAT, commissioning, qualification and handover documentation, and resolved faults raised during testing and site support.
- Integrated field devices, sensors, drives and process equipment with control logic and production data flows, with attention to diagnostics, safety, usability and data integrity.

Product Development Test Engineer (Contract) | Ford Motor Company via Invenio

Oct 2025 - Jan 2026

Little River, VIC

- Validated vehicle software integration and ADAS features in the Feature Software Integration and Validation team across the T6 Ranger and Everest programmes.
- Executed feature vehicle, breadboard and regression testing for readiness milestones, running changes, OTA updates and sign-off evidence across multiple vehicle variants.
- Instrumented test vehicles and conducted structured test drives under corporate test procedures in controlled and real-world conditions.
- Captured and analysed CAN bus data with Vector CANoe, CANalyzer and diagnostic tools for fault isolation, verification and evidence-based defect reporting.

Technical Assistant | ABMARC**Jul 2024 - Aug 2025***South Geelong, VIC*

- Conducted vehicle emissions and compliance testing against Australian Design Rules (ADR) and EURO standards across multiple vehicle platforms.
- Operated emissions test equipment, instrumentation and data acquisition systems to deliver repeatable, auditable and defensible results.
- Prepared compliance, QA and regulatory documentation for certification, audits and reporting, and analysed test data to identify deviations and trends.

Consultant Engineer - Internet of Things | DuxTel Pty Ltd**May 2024 - Aug 2024***Greater Geelong, VIC*

- Designed and deployed end-to-end IoT solutions linking sensors, embedded devices, gateways, cloud platforms and data pipelines to user-facing dashboards.
- Built real-time data ingestion, monitoring and visualisation with InfluxDB and Grafana, and delivered documentation and post-deployment support for client handover.

IoT Projects Administrator | DuxTel Pty Ltd**Feb 2024 - May 2024***Greater Geelong, VIC*

- Commissioned IoT devices onto ChirpStack-based LoRaWAN networks for low-power wide-area deployments across smart city, asset tracking and safety use cases.
- Developed an ESP32-based GPS tracking system with environmental sensing, covering firmware configuration, device provisioning, testing and deployment workflows.

Production Worker and Cellar Hand | IDL**Aug 2022 - Feb 2024***Moorabool, VIC | Casual role alongside full-time engineering study*

- Operated and monitored food and beverage production equipment, maintained throughput and quality, completed scheduled checks and reported faults with traceability records.

Production Line Operator | Carbon Revolution**Sep 2021 - Dec 2021***Waurin Ponds, VIC | Casual role alongside full-time engineering study*

- Set up and operated manufacturing equipment for high-performance carbon fibre wheels, maintaining quality, safety, routine maintenance and first-level troubleshooting standards.

Undergraduate Engineer | Thornton Engineering Australia Pty Ltd**Mar 2018 - Mar 2020***Corio, VIC*

- Supported engineering and QA for structural steel fabrication delivered to Australian Standards and client specifications.
- Prepared Inspection Test Plans (ITPs), Manufacturer Data Reports (MDRs) and quality records, and reviewed drawings for inspection planning, material traceability and compliance.

SELECTED ENGINEERING PROJECTS

- Autonomous Navigation Rover (ROS 2): Developed autonomous navigation in ROS 2 with Gazebo and RViz simulation, LiDAR-based SLAM, A* path planning, Kalman/EKF state estimation, and odometry and IMU sensor fusion.
- ESP32 Clinical Ataxia Assessment Device: Built a wire maze position tracking system for ataxia assessment using IMU, ToF, Hall-effect and magnetometer sensing on ESP32, with embedded firmware, MATLAB data logging and measurement validation.
- IoT GPS and Environmental Monitoring Platform: Delivered ESP32 GPS and environmental sensing with LoRaWAN and MQTT telemetry, ChirpStack device configuration, InfluxDB storage, Grafana dashboards and deployment documentation.

EDUCATION

Bachelor of Mechatronics Engineering (Honours) | Deakin University, Geelong VIC**Mar 2018 - Jul 2025***Graduated with Distinction***Higher National Diploma, Mechatronics, Robotics and Automation Engineering****Sep 2014 - Sep 2016***Cardiff Metropolitan University, UK**Graduated with Distinction***MEMBERSHIPS, CERTIFICATIONS AND LANGUAGES**

- Member, Engineers Australia | Current Victorian Driver Licence
- Lean Six Sigma Foundation
- JIRA and Agile
- KAIZEN
- Industrial Automation and IoT Foundations
- Languages: English, Tamil and Sinhala